

# Mathematical Foundations of Political Methodology Lecture 2

Robert Gulotty

Department of Political Science  
University of Chicago

August 28th, 2017

## Recap/motivation

Claim: The empty set  $\emptyset$  is a subset of every set.

Direct proof:

Recall “subset”,  $B \subset A$ , is *defined* to mean:

“For all  $x$  in the set  $B$ , those  $x$  are also in the set  $A$ .”

$$\forall x \in B, \quad x \in A$$

“The empty set  $\emptyset$  is a subset of every set.” can be rewritten,

$$\forall A, \quad \forall x \in \emptyset, \quad x \in A$$

Pick an arbitrary set  $A$ . We then test whether or not those  $x \in \emptyset$  satisfy  $x \in A$ . But we know that there are no  $x \in \emptyset$ .

$$\forall x \in \emptyset, \quad x \in A, \quad \square$$

# Why Proofs?

- Proofs are just arguments that use logical reasoning.
- Logical reasoning is a skill obtainable via practice and study.
- Formal theory can remove ambiguity and imprecision, creating possibility for surprising discoveries: Arrow Impossibility Theorem.
- Today we will focus on demystification and vocabulary.

# Today's plan

- Logical Notation
- Methods of Mathematical Proof
- Social Scientific Thinking

# Proposition

## Definition

A **proposition** is a statement that is either true or false.

Three examples:

Proposition:  $2 + 3 = 5$ .

Proposition: Oranges are a kind of vegetable.

Proposition: Superstition is the belief in the causal nexus.

# Proposition

In English, we can modify, combine, and relate propositions with words such as “not”, “and”, “implies”, and “if-then”. Such propositions are called as compound propositions.

| English   | In R     | Symbols       |
|-----------|----------|---------------|
| “not”     | ! (Bang) | $\sim$        |
| “and”     | &        | $\cap$        |
| “or”      | (Pipe)   | $\cup$        |
| “if-then” | ifelse   | $\rightarrow$ |

To understand compound propositions, we can use *truth tables*.

## Proposition: $\text{Not}(P)$

The truth table for " $\text{Not}(P)$ " is:

| $P$ | $\text{Not}(P)$ |
|-----|-----------------|
| T   | F               |
| F   | T               |

## Proposition: P and Q

The truth table for "P and Q" is:

| P | Q | P and Q |
|---|---|---------|
| T | T | T       |
| T | F | F       |
| F | T | F       |
| F | F | F       |

## Proposition: $P$ or $Q$

The truth table for “ $P$  or  $Q$ ” is:

| $P$ | $Q$ | $P$ or $Q$ |
|-----|-----|------------|
| T   | T   | T          |
| T   | F   | T          |
| F   | T   | T          |
| F   | F   | F          |

## Proposition: Implication ( $P \rightarrow Q$ )

In an 'if then' statement, we say the antecedent  $P$  implies the consequent  $Q$ .

The truth table for "P implies Q" is:

| P | Q | P implies Q |
|---|---|-------------|
| T | T | T           |
| T | F | F           |
| F | T | T           |
| F | F | T           |

In words:

- An implication is true exactly when the if-part is false or the then-part is true.
- In words: An implication is false exactly when the if-part is true and the then-part is false.
- "If my grandmother had wheels then she would have been a bike."

## Proposition: $P \text{ IFF } Q$

The truth table for  $P \text{ IFF } Q$  (if and only if)  $Q$  is:

| P | Q | $P \text{ IFF } Q$ |
|---|---|--------------------|
| T | T | T                  |
| T | F | F                  |
| F | T | F                  |
| F | F | T                  |

Example. This if-and-only-if statement is true for every real number  $x$ :  $x^2 - 4 \geq 0$  iff  $|x| \geq 2$

# Proposition: Contrapositive

## Definition

*“Not(Q) implies Not(P)” or  $\sim Q \rightarrow \sim P$  is called the **contrapositive** of the implication “P implies Q”.*

Their truth table is:

| P | Q | $P \rightarrow Q$ | $\sim Q$ | $\sim P$ | $\sim Q \rightarrow \sim P$ |
|---|---|-------------------|----------|----------|-----------------------------|
| T | T | T                 | F        | F        | T                           |
| T | F | F                 | T        | F        | F                           |
| F | T | T                 | F        | T        | T                           |
| F | F | T                 | T        | T        | T                           |

Example. “If I am hungry, then I am grumpy.” is equivalent to “If I am not grumpy, then I am not hungry”.

## Proposition: Always vs. Sometimes True

Examples on "Always True":

For all  $x \in \mathbb{R}$ ,  $x^2 \geq 0$ . Equivalent to write as:  $\forall x \in \mathbb{R}, x^2 \geq 0$ .

Examples on "Sometimes True":

There exists an  $x \in \mathbb{R}$  such that  $5x^2 - 7 = 0$ . Equivalent to write as:  $\exists x \in \mathbb{R}$  such that  $5x^2 - 7 = 0$ .

- Logical Notation
- **Methods of Mathematical Proof**
- Social Scientific Thinking

# Mathematical Proofs

Before we proceed to the definition of mathematical proofs, we need to define axioms first.

## Definition

*Propositions that are simply accepted as true are called **axioms**.*

Example of an axiom: There is a straight line segment between every pair of points.

# Mathematical Proofs

## Definition

A **mathematical proof** of a proposition is a chain of logical deductions leading to the proposition from a base set of axioms.

# Classes of Results

The importance of a proposition is indicated with its classification:

- Very important propositions are called **theorems**.
- A **lemma** is a preliminary proposition useful for proving later propositions.
- A **corollary** is a proposition that follows in just a few logical steps from a lemma or a theorem.

# Methods for Mathematical Proofs

A fundamental logical deduction is *modus ponens*. The truth of  $P$  and  $P \rightarrow Q$  gives us the truth of  $Q$ .

$$\frac{P, \quad P \rightarrow Q}{Q}$$

An example of an argument that fits the form *modus ponens*:  
If there is a skateboard on the sidewalk, my dog will bark at it.  
There is a skateboard on the sidewalk.  
Therefore, my dog will bark.

## Source of *modus ponens*

| P        | Q        | P implies Q |
|----------|----------|-------------|
| <b>T</b> | <b>T</b> | <b>T</b>    |
| <b>T</b> | <b>F</b> | <b>F</b>    |
| <b>F</b> | <b>T</b> | <b>T</b>    |
| <b>F</b> | <b>F</b> | <b>T</b>    |

# Classic Logical Fallacy

The following error in logic is called ‘affirming the consequent’:

I assume “If there is a skateboard on the sidewalk, my dog will bark at it.”

and see “My dog is barking.”

and erroneously conclude: “Therefore, there is a skateboard on the sidewalk.”

$$\frac{Q, \quad P \rightarrow Q}{P \vee \sim P}$$

## Source of affirming the consequent

| P | Q        | P implies Q |
|---|----------|-------------|
| T | <b>T</b> | <b>T</b>    |
| T | F        | F           |
| F | <b>T</b> | <b>T</b>    |
| F | F        | <b>T</b>    |

# Methods for Mathematical Proofs: Direct Proof

So far we have shown that if we believe  $P \rightarrow Q$ , and  $P$ ,  $Q$  follows.  
Most political science claims are like  $P \rightarrow Q$ , so how do we prove this?  
The most common form of proof is the *Direct Proof*

For this method, in order to prove that  $P \rightarrow Q$ , there are two steps: 1. Assume  $P$  is true; 2. Show that  $Q$  logically follows, usually from the definition of  $P$ .

## Source of *Direct Proofs*

| P        | Q        | P implies Q |
|----------|----------|-------------|
| <b>T</b> | <b>T</b> | T           |
| <b>T</b> | F        | F           |
| F        | T        | T           |
| F        | F        | T           |

# Methods of Mathematical Proof: Direct Proof

## Theorem

If  $0 \leq x \leq 2$ , then,  $-x^3 + 4x + 1 > 0$ .

## Proof.

Assume  $0 \leq x \leq 2$ . Then,  $x$ ,  $2 - x$ , and  $2 + x$  are all nonnegative. Therefore, the product of these terms is also nonnegative. Adding 1 to this product gives a positive number, so:

$$x(2 - x)(2 + x) + 1 \geq 0$$

Which is equivalent to:

$$-x^3 + 4x + 1 > 0$$



# Methods of Mathematical Proof: Direct Proof

Now, let's see an example of using direct proof in set theory.

## Theorem

*Let  $A$  and  $B$  be sets. If  $A = B$  then  $A \subset B$  and  $B \subset A$*

# Methods of Mathematical Proof: Prove the Contrapositive

Recall that  $\sim Q \rightarrow \sim P$  is the **contrapositive** of  $P \rightarrow Q$

The contrapositive is often easier to prove

To prove the contrapositive, there are two steps: 1. write, “We prove the contrapositive:”, and then state  $\sim Q \rightarrow \sim P$ ; 2. proceed as in the first method.

Now, let's see an example.

# Methods of Mathematical Proof: Prove the Contrapositive

## Theorem

*If  $r$  is irrational, then  $\sqrt{r}$  is also irrational.*

## Proof.

We prove the contrapositive: if  $\sqrt{r}$  is rational, then,  $r$  is rational.

Assume that  $\sqrt{r}$  is rational. Then, there exists integers  $a$  and  $b$  such that:

$$\sqrt{r} = \frac{a}{b}$$

Squaring both sides gives:

$$r = \frac{a^2}{b^2}$$

Since  $a \times a$  and  $b \times b$  are integers (integers are closed in multiplication):  
 $r$  is also rational. □

# Methods of Mathematical Proof: Prove by Contradiction.

In a proof by contradiction, you suppose that the thing you want to prove is not true. You then show that such a supposition contradicts other propositions.

The step of proof by contradiction is: 1. Write, "We use proof by contradiction."; 2. Write, "Suppose  $P$  is false"; 3. Deduce something known to be false (such as a logical contradiction); 4. Write, "This is a contradiction. Therefore,  $P$  must be true."

# Methods of Mathematical Proof: Proof by Contradiction.

## Theorem

*There is no largest even integer.*

## Proof.

Suppose that there is a largest even integer, its name is  $k$ .

Since  $k$  is even, it has the form  $2n$ , where  $n$  is also an integer. Integers are closed under addition. Consider  $k + 2$ .  $k + 2 = (2n) + 2 = 2(n + 1)$ .

So  $k + 2$  is even.

But  $k + 2$  is larger than  $k$ . This contradicts our assumption that  $k$  is the largest even integer  $\square$ .

# Methods of Mathematical Proof: Proof by Contradiction.

Now, let's see an example of using proof by contradiction in set theory.

## Theorem

*Let  $A$  and  $B$  be sets. Then  $A = B$  if and only if  $A \subset B$  and  $B \subset A$ .*

## Proof.

$\Rightarrow$  Suppose  $A = B$ . By definition, if  $x \in A$ ,  $x \in B$ . So  $A \subset B$ . Again, by definition, if  $y \in B$  then  $y \in A$ . So  $B \subset A$ .

$\Leftarrow$  Suppose  $A \subset B$  and that  $B \subset A$ . Now, by way of contradiction, suppose that  $A \neq B$ .  $A \neq B$  only if there is  $x \in A$  and  $x \notin B$  or a  $y \in B$  and  $y \notin A$ . But then, either  $A \not\subset B$  or  $B \not\subset A$ , contradicting our initial assumption. □

## An aside on Hypothesis Testing.

We must decide what to reject upon arriving at a contradiction. A related point occurs in hypothesis testing.

- We suppose some 'null hypothesis' is true: there is no effect of pasta on weight gain.
- We run an experiment.
- We find evidence that is unlikely to arise by chance if pasta does not affect weight gain and our experiment was well run.
- Do we conclude that pasta causes weight gain or that our experiment was not well run?

## Proof by Induction.

We want to show that a property holds for all  $n$ , where  $n$  is an integer.

- 1 Prove it holds for  $n = 0$  (or  $n = 1$ ).
- 2 *Assume* it holds for arbitrary  $n$ .
- 3 Prove it must hold for  $n + 1$ .

## Example of proof by Induction.

Claim: If  $|X| = n$  then  $|2^X| = 2^n$  (The size of the power set of a set of size  $n$  is  $2^n$ )

**1** Prove it holds for  $n = 0$ :

- Suppose that  $|X| = 0$ .
- If  $X$  has no elements, then  $X = \emptyset$ .
- It follows that  $2^{\{\}} = \{\{\}\}$ : the set of all subsets of  $\emptyset$  is just the set containing  $\emptyset$ .
- Thus,  $|2^{\{\}}| = 2^{|X|} = 2^0 = 1$ , proving our statement for  $n = 0$ .

**2** Suppose our statement holds for  $|X| = n$ , so  $|2^X| = 2^n$ .

**3** Prove it must hold for  $n + 1$ :

- Let  $Y = X \cup \{a\}$ , so that  $|Y| = n + 1$ .
- $2^Y$  is  $2^X \cup \{S : S \subset Y \text{ and } a \in S\}$ .

## Example of proof by Induction.

- Suppose  $X = \{1, 2\}$ ,  $Y = X \cup \{3\} = \{1, 2, 3\}$ .
- $2^X = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ .
- $2^Y = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, \{3\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$ .
- $2^Y = 2^X \cup \{S : S \subset Y \text{ and } 3 \in S\}$ .

## Example of proof by Induction.

Claim: If  $|X| = n$  then  $|2^X| = 2^n$  (The size of the power set of a set of size  $n$  is  $2^n$ )

1 Prove it holds for  $n = 0$ :

- Suppose that  $|X| = 0$ .
- If  $X$  has no elements, then  $X = \emptyset$ .
- It follows that  $2^{\{\}} = \{\{\}\}$ : the set of all subsets of  $\emptyset$  is just the set containing  $\emptyset$ .
- Thus,  $|2^{\{\}}| = 2^{|X|} = 2^0 = 1$ , proving our statement for  $n = 0$ .

2 Suppose our statement holds for  $|X| = n$ , so  $|2^X| = 2^n$ .

3 Prove it must hold for  $n + 1$ :

- Let  $Y = X \cup \{a\}$ , so that  $|Y| = n + 1$ .
- $2^Y$  is  $2^X \cup \{S : S \subset Y \text{ and } a \in S\}$ . Note that  $2^X$  and  $\{S : S \subset Y \text{ and } a \in S\}$  are disjoint.
- The set  $\{S : S \subset Y \text{ and } a \in S\}$  has  $2^n$  elements: it is  $\{S : S = \{x\} \cup Z \text{ where } Z \in 2^X\}$ .
- Thus,  $|2^Y| = 2^n + 2^n = 2 * 2^n = 2^{n+1}$ .

4 This proves our result.  $\square$

- Proposition
- Methods of Mathematical Proof
- Social Scientific Thinking

# Social Scientific Thinking

A *positive theory* in social science asserts that two or more concepts or events are related to each other and specifies the reason why.

Reasons why are called 'mechanisms'.

We may aim to explain (interpret) or predict.

# Internal Validity

## Definition

*In scientific research, **internal validity** pertains to whether the causal conclusion of a study is warranted.*

This is usually a question of research design and logical merit.

Example. If the experiment engages in proper randomization, it may have internal validity.

# External Validity

## Definition

**External validity** *pertains to the ability to generalize (causal) inferences in scientific research.*

*In other words, it is the extent to which the results of a study can be generalized to other situations and to other people.*

Example. If the randomized controlled trial is conducted on a representative sample, it may have external validity.

We may be concerned that our study of war from 1853-1856 doesn't travel to 264 BC.

# Hypothesis

## Definition

*In general a **hypothesis** is a statement made on the basis of evidence that can be tested. The purpose of a hypothesis is to organize a study.*

Example. A hypothesis can be: economic fluctuation causes civil conflict.

# Model

## Definition

*Scientists use the term 'model' to convey an implication of greater order and system in a theory.*

- A model represents a portion of reality, either an object or a process, in such a way as to highlight what are considered to be key elements or parts of the object or process and the connections among them
- Any representation of a system, whether in words or diagrams, is a model.

Example. The rational choice model of political candidates on one dimension.

# Stats/social science Terminology

## Definition

*We predict or explain dependent variables on the basis of an independent or explanatory variable.*

Example: Higher income causes the country to be democracy. The independent variable is per capita income. The dependent variable is whether or not the country is a democracy. If income is also caused by the extent of democracy in a country, we say income is 'endogenous.'

We often discuss variables in terms of a regression equation:

$$Y = \alpha + \beta X + \epsilon$$

You will hear people say 'left hand side' for the dependent variable ( $Y$ ) and 'right hand side' for the independent variable ( $x$ ).

## What are (multivariate) regressions for?

$$Y = \alpha + \beta_1 X + \beta_2 Z + \epsilon$$

$$\beta_1 = E[Y|X = x + 1, Z = z] - E[Y|X = x, Z = z]$$

- Summary: Regressing  $Y$  on  $X$  and  $Z$ , the coefficient of  $X$ ,  $\beta_1$ , is the expected difference in  $Y$  per difference in  $X$ , comparing pairs of items that differ in  $X$  but are identical in  $Z$ .
- Prediction: Regressions give us a model to predict some continuous outcome given linear extrapolations of our data,  $X$  and  $Z$ .
- Causation: Regressions give us a model to predict how the world would have been given different values of  $X$ , where  $Z$  is a potential confounding factor associated with  $X$  and  $Y$ .

## Instructions going forward

- rework through Pemberton & Rau Ch. 5
- Finish homework.